

# Real-World Data Cleaning

## 1 The Reality of Real-World Data

In theoretical exercises, data is often assumed to be pristine and perfectly formatted. In real-world applications, however, data is typically collected from human subjects or imperfect sensors, leading to a variety of errors, typos, and omissions. Data cleaning is not merely a preliminary administrative chore; it is a critical analytical step.

Statistical algorithms and probability models are highly sensitive to input data. If poorly formatted or erroneous data is fed into a model, the output will be mathematically sound but practically meaningless—a principle commonly known as “garbage in, garbage out” (GIGO).

Before any formal probability calculations or statistical tests can be performed, a dataset must be carefully audited. In this chapter, we formalize the policies for handling two primary classes of data imperfections: structural inconsistencies and missing values.

## 2 Structural Inconsistencies and Categorical Recoding

Structural inconsistencies arise when the same underlying logical truth is recorded using multiple different formats. This is particularly common in free-text survey fields.

Consider a simple binary question asking subjects whether they play video games. The intended sample space for the event is  $S = \{\text{Yes}, \text{No}\}$ . However, human respondents might enter: “Yes”, “yes”, “Yeah”, “yep”, “Y”, “No”, “nope”, or “N”.

Computational algorithms do not inherently understand synonyms; they treat every unique string of characters as a strictly distinct category. To analyze this data, we must standardize it through a process called *categorical recoding*.

**Definition 1** (Categorical Recoding). *Categorical recoding (or mapping) is the systematic process of converting text variations and inconsistent data entries into a uniform set of numerical or standardized categorical variables.*

The standard policy for categorical recoding involves three steps:

1. **Extraction:** Generate a complete list of every unique response present in the dataset.
2. **Dictionary Definition:** Define a strict numerical dictionary. For binary events, it is standard practice to map the absence of the trait to 0 and the presence to 1 (e.g., 0 = No, 1 = Yes). For multi-class variables, use sequential integers (1, 2, 3, ...).
3. **Mapping:** Manually group the raw text variants and assign them to the predefined numerical keys. For example, the subset {"Yes", "yeah", "yep"} is mapped entirely to the integer 1.

Once recoded into numerical indicators, the data is ready for the rigorous probability operations discussed in Chapter 1.

### 3 Handling Obvious Typos and Anomalies

Human error often yields data points that are physically impossible or structurally incorrect based on the requested units of measurement. These anomalies must be addressed before proceeding.

#### The “Obvious Intent” Rule

When an anomalous value is encountered, the analyst must decide whether the true value can be logically deduced. We govern this process using the “Obvious Intent” rule, which dictates two distinct paths based on the ambiguity of the error:

- **Case A: Unambiguous Intent.** If the nature of the error is mathematically obvious, the data point may be corrected. For example, if a column asks for a subject’s height in centimeters, and a value of 1.8 is found alongside values like 175 and 182, it is logically certain that the respondent entered their height in meters. The analyst may safely multiply the anomaly by 100 to correct it to 180.
- **Case B: Ambiguous Intent.** If the true intent cannot be deduced with absolute certainty, the data point must not be guessed. For example, if a university student’s age is entered as 2, it is impossible to know if they meant 20, 21, or 22. In this scenario, the erroneous value must be deleted and converted into a formally *missing value* (often denoted as NA or Null).

### 4 Missing Data Policies: Drop or Impute?

After mapping categories and stripping ambiguous typos, the dataset will inevitably contain missing values. How an analyst handles these gaps profoundly dictates the integrity of the subsequent statistical tests.

We broadly categorize the handling of missing data into three hierarchical approaches.

## 4.1 Pairwise Deletion (Available Case Analysis)

If an analyst is calculating a specific joint probability or a statistic (such as covariance) between random variables  $X$  and  $Y$ , they only require data for  $X$  and  $Y$  to be present for a given trial. If a third variable  $Z$  is missing for that trial, it has no bearing on the calculation between  $X$  and  $Y$ .

- **Policy:** Exclude a trial (row) from calculation only if it is missing the specific variables required for the current mathematical operation.
- **Usage:** This is the standard, preferred approach in basic statistics, as it preserves the maximum amount of recorded data.

## 4.2 Listwise Deletion (Complete Case Analysis)

Listwise deletion is the blunt enforcement of data completeness. Under this policy, an entire trial (a row of subject data) is completely discarded if even a single variable within that trial is missing.

- **Policy:** Drop the entire row if there exists any missing cell.
- **Usage:** This approach is highly discouraged unless the proportion of missing data is mathematically negligible (e.g., discarding 5 rows from a sample space of  $n = 10,000$ ). Otherwise, it severely shrinks the sample size and wastes valid data.

## 4.3 Data Imputation

When data cannot be safely dropped, analysts may choose to fill in the gaps.

**Definition 2** (Imputation). *Imputation is the statistical procedure of replacing missing data with substituted values, which are calculated or deduced from the known values within the dataset.*

Elementary imputation relies on measures of central tendency:

- **Mean Imputation:** Replacing missing values with the mathematical average of the available data in that column. Best suited for continuous, normally distributed data.
- **Median Imputation:** Replacing missing values with the median. Best suited for skewed continuous data or data containing extreme outliers.

- **Mode Imputation:** Replacing missing values with the most frequent response. Strictly used for categorical variables.

### The Danger of Imputation

Imputation is a mathematical artificiality. By definition, replacing missing values with a measure of central tendency artificially reduces the variance (spread) of the dataset.

For instance, imputing 50 missing weights with the sample mean will create a massive, unnatural spike in the probability distribution exactly at the mean. This distortion can invalidate subsequent probability calculations.

**Rule of Thumb:** Imputation should generally be restricted to variables where less than 5% to 10% of the data is missing. If a variable is missing a significant portion of its data, imputation is mathematically dangerous, and the variable itself should be dropped from the study.